

STEP 1 Score blocks

1	2	3	4	5	6	7	8
x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8

blocks

$e_{n,k}(a,b)$

$k=1$ $k=2$ $k=3$

[1:2]	-1.20	-0.30	-2.10
[3:5]	-0.70	-1.10	-0.40
[5:6]	-1.50	-0.20	-1.00
[6:8]	-0.60	-0.80	-0.10

STEP 2 Forward DP

$\alpha_n(j,k)$: mass ending at j with skill k

	0	1	2	3	4	5	6	7	8
start	1								
$k=1$									
$k=2$									
$k=3$									

arrows append blocks, weighted by $e_{n,k}(a,b)$

$$Z_n(U) = \sum_{S_n, z_n} p(S_n, z_n) \prod_{\ell} e_{n, z_{\ell}}(a_{\ell}, b_{\ell})$$

STEP 3a Sample a path

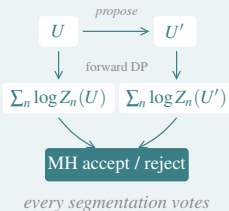
where do skill instances begin and end?

$z=2$		$z=1$			$z=3$		
x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8

drawn 3rd drawn 2nd drawn 1st

backward FFBS: sampled right to left

STEP 3b Path-marginal



total mass: the coloured path and all others